# TorqBall — Ball & Boards Test

*2026-09-06. The prototype **gate** for TorqBall — the same role the [[TorqSiege — Collapse Test Protocol v1]] plays for TorqSiege and [[TorqCapture — Zone-Capture Tech]] plays for TorqCapture. TorqBall's whole premise ([[TorqBall — Design Study v1]]) rests on two physical unknowns: which ball a speed-capped ~1 kg car can actually play, and what board build keeps ball + cars in and rebounding cleanly. This doc states both gating questions, analyses ball candidates with honest INR prices, gives a cheap test protocol + pass/fail bar, specs the boards build with a BOM, and gives a bench/rehearsal test for the boards. A third, related gate — the [[TorqBall — Boost & Attachments]] Grabber mechanic — gets its own bench test in **Part C** below. **Don't build the full arena until Parts A + B pass; don't build a Grabber head until Part C passes.** Prices from a Sept-2026 India-market scan (sources at the bottom).*

## The two questions we're answering

Not "can a car push a ball" (yes, trivially) and not "can plywood stand up" (also yes). The real bars:

1. **The ball:** which candidate ball does a speed-capped ~1 kg car actually *play well* — visibly shoved, riding over the bumper not under it, controllable enough for a real match, not wedging or dying after one hit?
2. **The boards:** what panel height/material/facing/corner design keeps the ball and the cars inside a boarded pitch, rebounds them cleanly off the walls (rebounds are the game), survives a car nosing into it repeatedly, and doesn't snag or injure?

Nail both and the rest of the rulebook is easy; miss either and no ruleset saves it.

---

## Part A — the ball test

### Candidate balls

**Size ceiling, stated up front:** the locked spec ([[TorqBall — Rulebook]] §2–§4) is a **~40–60 cm ball inside 35–45 cm boards**, with the boards' own containment rule at **board height ≥ ~60% of ball diameter** (§Part B below). A 45 cm board caps out at a ~75 cm ball on that rule alone — but the rulebook's ball spec is already capped at 60 cm, and the board height was fixed at 35–45 cm partly so an overhead cam and a standing spectator still see the ball and scrum over the top; a ball anywhere near 75–90 cm defeats that sightline even where the 60%-rule technically "allows" it. So the candidate list below is **capped at the Rulebook's own 60 cm ceiling** — nothing larger is worth testing, because even a pass would produce a ball the locked boards can't contain without either exceeding their spec or beating their purpose.

| Candidate | Size | Material | Glow option | ~Rs | Notes |
|---|---|---|---|---|---|
| **Stability/gym ball ("yoga ball"), small-diameter SKU** | 55–60 cm (buy the smaller SKU, not the 65/75 cm size) | PVC, anti-burst | LED insert kit (aftermarket) | 500–1,200 | Firmer skin than a beach ball, holds shape under a shove better, still light — the top of the tested range |
| **Foam-covered training ball (oversized)** | 40–50 cm | Dense foam shell over a light core | Painted with UV/glow paint, lit from a small internal LED puck | 500–1,200 | Smallest of the set — closest to the risk of wedging under a chassis; test the clearance explicitly |
| **Space-hopper / hop ball, small SKU** | 45–60 cm (avoid 65 cm+ SKUs) | Thick PVC, has a raised handle nub | Battery LED insert | 400–900 | The handle nub is a snag risk against the boards — flag it in testing |
| **Mid-size PVC beach ball, small SKU** | 50–60 cm only (NOT the common 60–90 cm party SKU) | Vinyl/PVC | Battery LED flicker insert, or a glow-stick taped inside | 200–500 | Cheap, light; the most likely to "ping around uncontrollably" per the design study's own risk — test first. Confirm the actual inflated diameter before buying — most beach-ball listings are sized for the 60–90 cm party market and oversell what a 60 cm SKU actually inflates to |
| **Light-up novelty beach ball (pre-lit), small SKU** | 55–60 cm only | Vinyl, built-in battery LEDs | Built-in — no rigging needed | 500–1,200 | Sold pre-lit on Amazon/Flipkart around festival season; most festival SKUs run 60–90 cm — check the listing's stated diameter, don't assume |

**Dropped from the shortlist (2026-09-06 fix):** the common 60–90 cm party/novelty beach-ball and gym-ball SKUs are **excluded**, not tested. At 75–90 cm they'd need 45–54 cm boards under the 60%-rule — at or past the 35–45 cm spec ceiling, and tall enough to block the overhead-cam/standing-spectator sightline the board height was set to protect. Buying and testing an oversized ball risks "winning" a candidate the locked boards physically can't contain, so it's cheaper to exclude the size band up front than to re-test after a false pass.

Buy **3–4 of the above**, not all five — pick a spread across size/material (e.g. one small-SKU beach ball, one gym ball, one foam ball, one pre-lit novelty ball) so the test actually compares different failure modes, not five variants of one. **At purchase, measure the actual inflated diameter with a tape measure before running any test** — retail listings for "beach ball" and "novelty ball" SKUs are inconsistent and often describe the 60–90 cm size even when a smaller variant exists.

## The test protocol (a kitchen-table afternoon, near-zero cost)

**What you need:** 1 speed-capped rented-class car (or any RC buggy you can get today — borrow, don't wait for the fleet), 3–4 candidate balls, a phone for video, a tape measure, a flat ~4×4 m space, a notebook.

**Method — run every ball through the same five checks, film each:**

1. **Push test.** Drive the car straight at the stationary ball from ~1 m. Does it move the ball cleanly, or does the ball absorb the hit and barely shift?
2. **Control test.** Dribble the ball in a straight line for ~3 m, then try a turn. Does it track predictably in front of the bumper, or does it ping off unpredictably?
3. **Ride-over test.** Nose the car directly into the ball at an angle. Does the ball ride up and over the bumper (good), or does the car's nose dip under it and stall (bad — the wedging failure mode)?
4. **Durability test.** 10 repeated hits at capped speed. Does the ball hold its shape/inflation, or does it deform, puncture, or the glow rig fail?
5. **Visibility test.** Film from directly overhead (a chair/ladder/phone mount) and from the side, in the venue's likely lighting. Does the ball read clearly on camera against the turf/boards?

**Score each ball 1–5 on three axes and log it:**

| Axis | 1 | 3 | 5 |
|---|---|---|---|
| Control | Pings away, uncontrollable | Movable but unpredictable | Shoves and dribbles cleanly, tracks in front of the bumper |
| Visibility | Disappears on camera / no glow | Visible but dull | Reads huge, glow pops on camera |
| Not-wedging | Wedges under the chassis and stalls the game | Occasionally catches, recovers with a nudge | Always rides over the bumper, never wedges |

Log: candidate, all three scores, durability notes (deflated/punctured/held up), and any snag risk (handle nubs, seams).

### Pass / fail bar

- **PASS** = at least one candidate scores **4–5 on all three axes** (control, visibility, not-wedging) and survives the 10-hit durability round without deflating or losing its glow. → Lock that ball, buy 3–4 more as event spares, freeze its spec into [[TorqBall — Rulebook]] §4 and cost it into the event needs list.
- **CONDITIONAL** = the winner needs a constraint (e.g. only plays well below a certain inflation pressure, or the glow rig needs a specific mount to survive hits). → Buildable, but bake the constraint into the pit's spare-ball prep routine.
- **FAIL** = nothing hits 4–5 across all three. → Don't build the full boarded arena yet; try a different size or material band **within the 40–60 cm ceiling** (going bigger is not an option — see the size-ceiling note above; a ball the boards can't contain isn't a usable win) or a hybrid (foam-covered inflatable) and re-test before committing. If nothing in 40–60 cm passes, the fallback is to revisit the board height (Part B) rather than the ball ceiling — but that reopens the sightline trade-off, so treat it as a last resort.

---

## Part B — the boards test

### The build

Modular dasher-board panels, designed to bolt/clamp together into a perimeter, break down flat for storage/transport, and survive repeated car hits without splintering or shifting.

- **Panel: 6–9 mm plywood**, cut to **~1.2 m wide × 0.35–0.45 m tall** modules. Plywood over MDF (MDF crumbles under repeated impact and hates any moisture).

- **Facing: EVA foam sheet (10–20 mm)** glued/velcroed to the impact face — softens car-to-board and ball-to-board hits, protects the plywood edge, and gives the ball a forgiving rebound rather than a hard bounce-back.
- **Height: 35–45 cm** — tall enough to contain the winning ball (Part A) and a shoved car, short enough that an overhead camera and a standing spectator still see the ball and the scrum over the top. Confirm the exact height against the winning ball's diameter once Part A locks (target: board height ≥ ~60% of ball diameter so the ball can't simply roll over the top on a hard hit).
- **Corners: radius corner panels** (curved plywood, or a straight corner wrapped in extra foam/pool-noodle padding) — no square corner. A square corner is where a ball jams and where a car chassis snags.
- **Base/feet:** each panel stands on a simple **plywood foot bracket or a sandbag-weighted base** — no floor drilling (venue floors are rented). Panels connect edge-to-edge with **carriage bolts + wingnuts** or **furniture-grade barrel bolts** for tool-free assembly/breakdown.
- **Goal openings:** two panels per short end are cut out to the goal spec ([[TorqBall — Rulebook]] §3) instead of solid — same plywood + foam edge treatment, with a shallow net or foam backstop just behind the opening.

### BOM (per ~10 × 6.5 m boarded pitch, ~33 linear metres of perimeter)

| Item | Qty — why | Est Rs | Notes |
|---|---|---|---|
| Plywood sheets (6–9 mm, 8×4 ft) | ~8–10 sheets to cut ~28 standard panels (1.2 m wide) covering ~33 m perimeter minus goal openings | 14,000–22,000 | 8×4 ft sheet ≈ Rs 1,400–2,200 depending on grade (Sept 2026 India retail) |
| EVA foam sheet (10–20 mm, for facing) | ~33 linear m × 0.4 m height ≈ 13 sq m | 6,500–13,000 | Rs 500–1,000/sq m depending on thickness/density |
| Radius/padded corner panels (4) | 4 corners | 2,000–4,000 | Curved ply + extra foam wrap, or straight corner with heavier foam padding |
| Carriage bolts + wingnuts / barrel bolts (panel joints) | ~56–60 (2 per joint × ~28–30 joints) | 1,500–3,000 | Hardware store, bulk |
| Foot brackets / sandbag bases | ~28–30 (1 per panel) | 3,000–6,000 | Simple angle-bracket foot, or sandbags reused from the shared perimeter-barrier stock |
| Goal backstop net/foam (2 openings) | 2 | 1,000–2,500 | Shallow net or foam wall just behind each goal opening |
| Matte paint/finish on boards (contrast for the glow ball) | lot | 1,000–2,000 | Dark matte finish so the lit ball reads against the boards on camera |

| Item | Qty — why | Est Rs | Notes |
|---|---|---|---|
| LED goal-frame strip, driver + PSU (optional, content) — **2 goal openings** | ~4–7 linear m total (each ~1.2–1.5 m opening framed to ~35–45 cm board height ≈ ~2–3.5 m of strip per goal × 2 goals) | 3,200–6,500 | Derived, not a flat guess: 4–7 m ≈ **1–2 × 5 m WS2812B-class addressable strips** at the same ~₹2,400–2,600/5 m rate cited from [[TorqCapture — Zone-Capture Tech]] (₹2,400–5,200 for the strip alone) + a small ESP32/driver board (₹400–700) + a modest 5V supply (₹400–600). *(Fixed 2026-09-06 — the prior single ₹1,500–4,000 line bundled two full goal frames' worth of strip with the board paint and undercounted against this exact same TorqCapture per-5m rate, which the doc already cited but didn't apply.)* |
| Fasteners, glue, misc hardware | lot | 1,000–2,000 | |
| **Boards one-time build subtotal** | | **≈ 33,200–61,000** | One-time — panels are modular, reusable every TorqBall event, break down flat for storage/transport. Optional LED goal-frame content line (₹3,200–6,500) can be dropped for a bare-bones build: **≈ 30,000–54,500 without it** |

This is the **one real fabrication** TorqBall adds over the soft-barrier kit every other TorqWars game uses — a one-time carpentry build, not a per-event consumable. Once built, the boards are reused event to event like the rented fleet's foam bump-rings.

### The bench/rehearsal test for the boards

**What you need:** at least 4–6 assembled panels (enough to form one full wall + one corner, not the whole pitch), 1 car, the winning ball from Part A, a phone for video.

**Method:**

1. **Rebound test.** Push the ball into the board wall at a shallow angle and at a square angle. Does it rebound cleanly back into play, or does it die against the foam / roll along the base and stick?
2. **Car-impact test.** Drive the car into the board at capped speed, straight on and at an angle. Does the board hold its position (no shifting/tipping), and does the car bounce back without damage?
3. **Corner test.** Push the ball and the car into a corner panel repeatedly. Does the radius/padding prevent jamming, or does the ball or the car wedge into the corner?
4. **Snag/safety check.** Run a hand along every panel edge, joint and corner at car height. Any sharp edge, exposed bolt head, or gap a finger/chassis could catch in?
5. **Assembly/breakdown timing.** Time a 2-person crew assembling the 4–6 panel test wall from flat, and breaking it back down. Extrapolate to the full ~28–30 panel perimeter for the event build-morning schedule.

**Pass / fail:**

- **PASS** = the ball rebounds cleanly at both angles, the car bounces off without damage or board movement, no jamming at the corner, no snag point found, and the assembly rate projects to a buildable build-morning window. → Lock the panel spec, order the full ~28–30 panel run, cost it into the event needs list.
- **CONDITIONAL** = works but needs a tweak (heavier base weighting to stop shifting, thicker foam at the corner, a taller/shorter panel to match the final ball size). → Buildable, bake the constraint into the BOM before the full order.

- **FAIL** = the ball or car regularly jams/wedges, boards shift or tip under a car hit, or a real snag/safety issue is found. → Don't order the full run; revise the panel design (heavier base, different foam density, different corner geometry) and re-test.

---

## Part C — the Grabber bench test (added 2026-09-06)

The third gating unknown, alongside the ball and the boards: [[TorqBall — Boost & Attachments]]'s **Grabber** boost (a velcro-faced attachment that makes the ball stick so a car can dribble it) is the doc set's "standout mechanic," but it is untested — whether velcro reliably grabs on a glancing hit, releases cleanly on a rival's bump, and survives repeated contact without peeling a light inflatable ball's shell is a real physical question, not a certainty. This section gives it the same test-then-build treatment as Parts A and B.

**What you need:** the winning ball from Part A (plus one disposable/practice ball, since the patch is glued/stitched on and testing may damage it), 1 speed-capped car fitted with a velcro-faced Grabber head (a plow attachment with hook-side velcro), a matching loop-side velcro patch on the practice ball, a phone for video, a stopwatch.

**Method:**

1. **Grab-on-contact test.** Approach the ball at ~10 different angles (head-on, 30°, 60°, glancing) at capped speed. Does the velcro grab and hold on contact, or does it bounce off without sticking? Log a grab/no-grab per angle.
2. **Dribble-control test.** With the ball grabbed, drive a straight line for ~3 m, then attempt a turn. Does the car retain control of the ball through the turn, or does it detach unpredictably or drag the ball off-line?
3. **Clean-release test.** With the ball grabbed, have a second car bump the carrying car from the side/behind at capped speed. Does the ball release cleanly (so play continues fairly), or does it stay stuck, jam, or take an oddly-directed bounce?
4. **Durability test.** Repeat the grab/dribble/release cycle **10 times** on the same patch. Does the velcro patch stay adhered and functional, or does it peel, tear the ball's shell, or lose grip strength?

**Score and log:** grab rate (X/10 angles), dribble outcome (clean / drags / detaches), release outcome (clean / stuck / erratic) per bump, and patch condition after 10 cycles (intact / degraded / failed).

**PASS / CONDITIONAL / FAIL bar:**

- **PASS** = grabs on a clear majority of contact angles (≥7/10), dribbles a useful distance with at least one controllable turn, releases cleanly on a rival bump most of the time, and the patch survives 10 cycles without peeling or damaging the ball. → Build the Grabber head for the event per [[TorqBall — Boost & Attachments]] §2a, patch every match ball + spare.
- **CONDITIONAL** = grabs/dribbles well but doesn't release cleanly on a bump (stays stuck too long), or the patch degrades but doesn't fully fail by cycle 10. → Buildable with a constraint: add the hard ~15–20 s release timer (already speced as a fallback) and/or reinforce the patch (double-stitch, a stiffer backing) before the event.
- **FAIL** = grab rate is low, the ball detaches uncontrollably rather than dribbling, or the patch fails (peels, tears the ball) within 10 cycles. → Don't build Grabber for v1; rework the velcro grade/patch size/head shape and re-test, or drop Grabber from the launch boost set and revisit later.

---

## Do-it checklist

1. Buy 3–4 candidate balls today (Part A); run the push/control/ride-over/durability/visibility checks; score and log.
2. If PASS, lock the ball and freeze its spec into [[TorqBall — Rulebook]].
3. In parallel or right after, build 4–6 test board panels (Part B); run the rebound/car-impact/corner/snag/assembly checks against the winning ball.
4. If PASS, order the full ~28–30 panel run and cost the boards + the locked ball into the event needs list.
5. Report both winning specs (ball + panel design) back here before any full arena spend.
6. Separately (not gating the arena, but gating the Grabber boost), run Part C on the winning ball with a velcro-patched practice ball + a fitted car; report the PASS/CONDITIONAL/FAIL result into [[TorqBall — Boost & Attachments]] §2a before building any Grabber head for an event.

---

***Price sources (Sept 2026):*** *PVC beach balls and stability/gym balls — Amazon.in/Flipkart general listings, Rs 200–1,500 range depending on size and anti-burst rating; foam-covered training balls and space-hoppers — sports-goods retailers and Amazon.in, Rs 400–1,200; pre-lit novelty LED beach balls — seasonal festival-lighting sellers on Amazon.in/Flipkart, Rs 500–1,500. Plywood 6–9 mm 8×4 ft sheet — hardware/timber market retail, Rs 1,400–2,200/sheet; EVA foam sheet — craft/packaging suppliers, Rs 500–1,000/sq m depending on thickness; carriage bolts/wingnuts, brackets, sandbags — hardware store, standard retail. LED strip for goal frame — same WS2812B-class pricing as [[TorqCapture — Zone-Capture Tech]] (~Rs 2,400–2,600 per 5 m addressable strip).*

***Verified + changelog (2026-09-06):*** *first build, in the TorqSiege Collapse Test / TorqCapture Zone-Capture Tech "prototype gate" pattern. Stated both gating questions (ball control/visibility/not-wedging; board containment/rebound/durability). Part A: shortlisted 5 candidate ball types with size/material/glow options + honest Rs ranges, wrote a 5-check test protocol (push/control/ride-over/durability/visibility) with a 3-axis 1–5 scoring rubric and a PASS/CONDITIONAL/FAIL bar. Part B: specced the modular dasher-board build (6–9 mm plywood, EVA foam facing, 35–45 cm height tied to the winning ball's diameter, radius corners, tool-free bolt joints) with a full costed BOM (~Rs 30,500–56,500 one-time for a ~33 linear-metre perimeter) and a bench/rehearsal test (rebound, car-impact, corner-jam, snag/safety, assembly timing) with its own PASS/CONDITIONAL/FAIL bar. Both tests explicitly gate [[TorqBall — Rulebook]] — no full arena spend before both pass.*

***Second pass (2026-09-06, examiner fixes):*** *(1)* ***Ball size ceiling capped at ~60–65 cm*** *— the original shortlist ran candidates up to 90 cm ("40–90 cm range tested") against the locked 40–60 cm ball spec and 35–45 cm boards; under the doc's own 60%-of-diameter containment rule a 75–90 cm ball needs 45–54 cm boards, exceeding the board spec and defeating the overhead-cam/standing-spectator sightline the board height protects. Reworked the candidate table to small-diameter SKUs only, added an explicit "measure before buying" warning (retail listings for beach/novelty balls default to the 60–90 cm party size even when a smaller SKU exists), and stated the size-ceiling reasoning up front so it can't silently regress. (2)* ***Added Part C — the Grabber bench test*** *(grab-on-contact rate across 10 angles, dribble control, clean-release on a bump, 10-cycle patch durability) with its own PASS/CONDITIONAL/FAIL bar, gating [[TorqBall — Boost & Attachments]]'s Grabber build the same way Parts A/B gate the arena. (3)* ***Fixed the goal-frame LED under-costing*** *— the doc cited TorqCapture's ~₹2,400–2,600/5 m strip rate but then bundled two full lit goal frames' worth of strip into a single ₹1,500–4,000 "paint + LED" line; split paint (₹1,000–2,000) from* *a derived LED line (~4–7 m ≈ 1–2 × 5 m strips + driver + PSU, ₹3,200–6,500) and recomputed the boards subtotal end to end:* ***≈ ₹30,500–56,500** → **≈ ₹33,200–61,000*** *(≈ ₹30,000–54,500 with the optional LED line dropped).*

***Third pass (2026-09-06, examiner fixes):*** *the size ceiling drifted to "~60–65 cm" in the candidate-list intro and the FAIL-retry line, and the gym-ball candidate's own range ran to 65 cm — all inconsistent with the locked [[TorqBall — Rulebook]] §2–§4 spec of a* ***40–60 cm*** *ball. Capped all three to 60 cm: the intro now reads "capped at the Rulebook's own 60 cm ceiling," the gym-ball row is now "55–60 cm (buy the smaller SKU, not the 65/75 cm size)," and the FAIL bar's retry band is now "within the 40–60 cm ceiling." The rulebook's own spec is unchanged — this only tightens the candidate list and retry band to match it, since a 65 cm ball was never actually inside the locked spec being tested against.*